

Canadian Survey on Business Conditions

Analysis and Automated Quarterly Refresh

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Abstract

Four quarterly extracts of the Canadian Survey on Business Conditions (Q3 2023 – Q2 2024) are analysed, and a lightweight pipeline is built to absorb future quarters without manual rework.

Business expectations improved into Q2 2024, but the improvement is narrower than the national headline suggests. Sales, employment and profitability increase-expectations each reached four-quarter highs, while investment stayed flat. Two independent diagnostics show that investment is not a lagging indicator but a *separate decision*: one business in six gives no directional answer on it, and its expectations are statistically uncorrelated with the other three metrics. Beneath the national figure, 12 of 13 industries expect profitability to fall, and every industry expected it in each of the three preceding quarters, so margin pessimism is entrenched rather than newly emerging. It is easing (mean balance $-23.9 \rightarrow -17.1$ pp), but businesses have expected to sell more and earn less for a year, which is a cost-side rather than a demand-side story. The improvement is also regionally concentrated: Ontario and Quebec, the two largest economies, improved by less than the national average, and Manitoba declined.

Submission at a glance

Quarterly files analysed	4
Rows in combined dataset	9,408
Pieces of analysis	3
Missing VALUE cells identified	7
Automated tests	15

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1 The data

Each survey wave asks business owners across Canada whether they expect **Employment**, **Sales**, **Profitability** and **Investment** to *increase*, *stay about the same*, or *decrease* over the next three months. Results break down by geography and by business characteristic.

Field	Meaning
GEO	Geography, Canada or a province/territory
Business_characteristics	Grouping: all-industries aggregate, or one industry
Business_information	The metric asked about (one of the four above)
Expected_change	Direction: increase / stay about the same / decrease
VALUE	Percentage of business owners in that group giving that answer
Quarter	Survey quarter, e.g. Q2 2024

Table 1: Field definitions.

Each file is a complete grid: 14 geographies $\times$ 14 characteristics $\times$ 4 metrics $\times$ 3 directions = 2,352 rows, with four files giving 9,408 rows. Notably, only *one* of the 14 geographies is Canada, so a Canada-only analysis would leave about 93% of the rows untouched, which is why Section 6 uses them.

2 Data preprocessing

Implemented in `src/01_data_preprocessing.ipynb`. The guiding principle is to decide what *not* to assume.

2.1 Label drift

Q2 2024 renames two labels relative to earlier files. The mappings are explicit in code rather than fuzzy-matched, so a label nobody has mapped fails validation instead of quietly starting a second, parallel series.

Raw Q2 2024 label	Canonical label	Rows affected
Capital Investment	Investment	588
stay the same	stay about the same	784

Table 2: Known category drift, normalised on load.

2.2 Missing data

Seven VALUE cells are missing, all in Q2 2024. Counting them is not enough, their *position* identifies what kind of problem this is. All seven sit in the **first 18 rows of one file**, and **six of the seven are the decrease response**.

Survey suppression would scatter across small cells throughout a file. This is clustered at the top and concentrated in one response column, which reads as an export or copy artifact.

Decision: flag and retain, never impute. In particular the tempting reconstruction `decrease = 100 - increase - stay about the same` is invalid here, for the reason given in Section 3.1. Each analysis is then designed so its published figures depend only on observed values.

2.3 Abnormalities

Two things look wrong at first glance and need separating, because they call for different handling.

A. Empty cells. 122 groups have all three shares at exactly zero. These are *absent* cells: a group too small to report, not partial answers. They are counted separately, because averaging them in would drag every residual statistic toward zero and hide the real pattern.

B. Shares that do not sum to 100. Of 3,130 complete groups, 1,585 sum to approximately 100 and 1,423 fall materially short. This is the important one, and Section 3.1 shows it is not random.

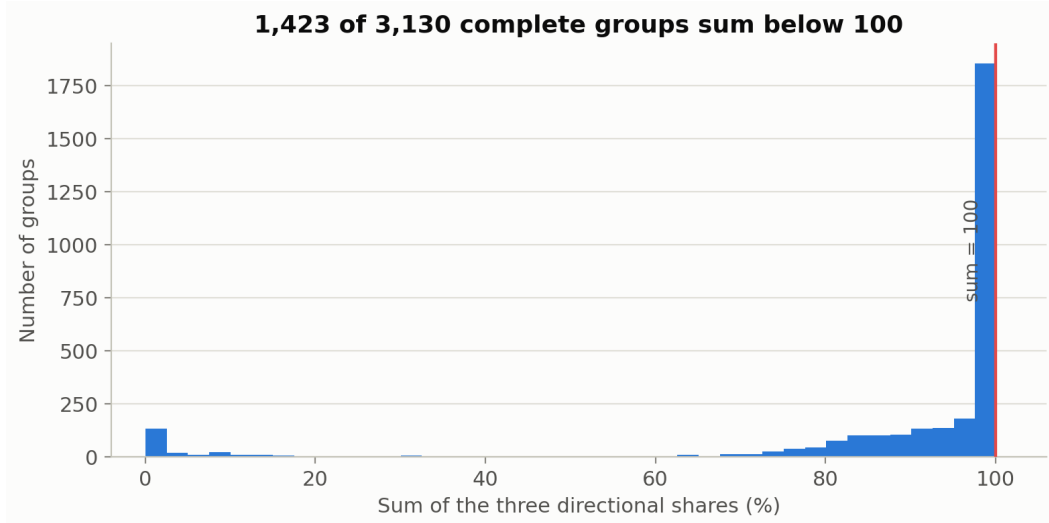


Figure 1: The spike at 100 is Employment and most of Sales and Profitability; the long left tail is almost entirely Investment. The small cluster at zero is the 122 empty cells.

2.4 Validation

Seven checks run on every file. Five **raise**; a structural problem must stop the load, and two **warn** and record a metric, because a characteristic of the data should be surfaced rather than hidden.

#	Check	On failure
1	All six required columns present	raise
2	Exactly one valid Q# YYYY per incoming file	raise
3	VALUE numeric and within 0–100	raise
4	No duplicate natural key	raise
5	Only known category labels after aliasing	raise
6	Each group has all three direction rows	warn + record
7	Missing percentages	warn + retain

Table 3: The validation contract. The natural key is GEO × characteristic × metric × direction × quarter.

No duplicate keys were found, and every observed percentage falls within 0–100. A machine-generated summary is written to `outputs/data_quality_report.txt`.

3 Understanding the data

Implemented in `src/02_understanding_data.ipynb`. Each question is paired with the chart form that answers it.

3.1 How do responses divide within a question?

Form: donut: four segments that genuinely make a whole, which is the one case where a part-to-whole circle beats a bar. The fourth segment is the share giving *no directional answer*: the gap left when the three published shares do not reach 100.

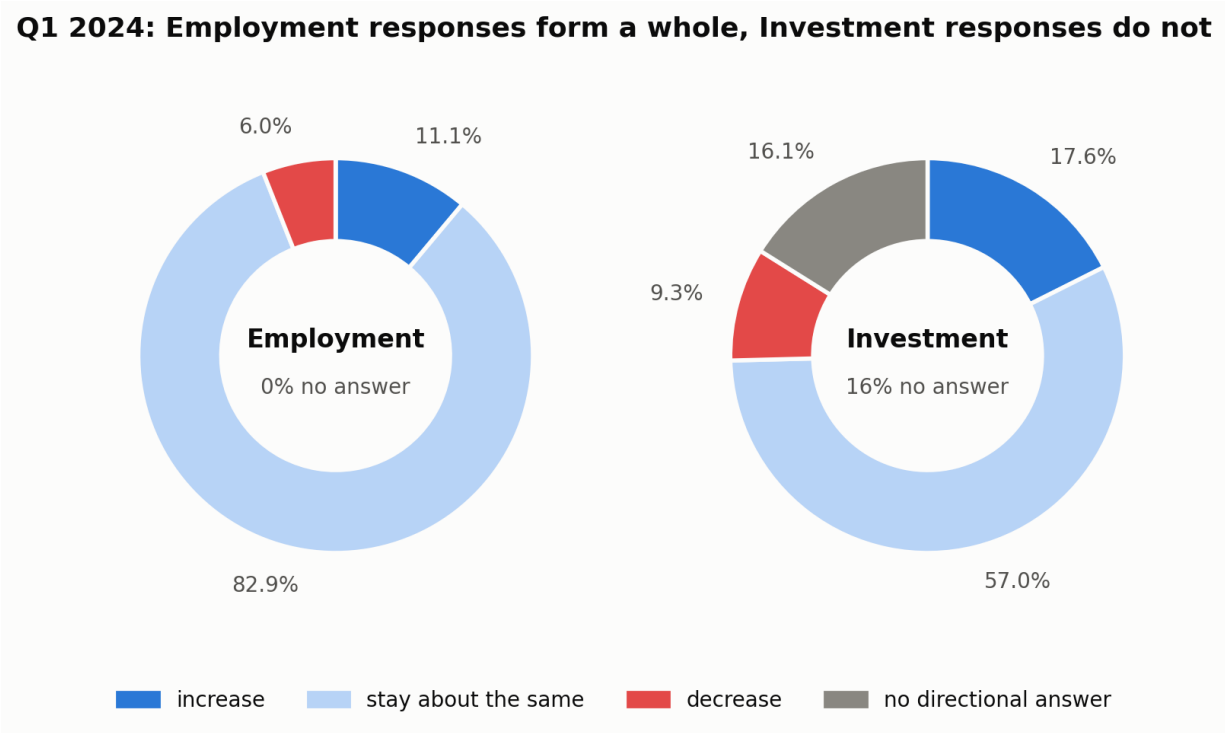


Figure 2: Q1 2024, Canada, all industries. Employment’s responses account for every business; Investment’s leave 16% unaccounted for.

Employment’s three shares sum to exactly 100. Investment’s sum to about 84, roughly one business in six gives no directional answer on investment at all.

The likeliest explanation is mundane and important: many businesses, especially small ones, have no investment plan to report, so there is no direction to give. Employment is something every operating business can answer.

Metric	Mean residual (all cells)	Canada, all industries		
		Q3 2023	Q4 2023	Q1 2024
Employment	2.2	0.0	0.0	0.0
Profitability	6.5	2.9	2.7	2.6
Sales	6.4	3.4	3.4	3.8
Investment	18.5	15.3	14.9	16.1

Table 4: The unexplained residual (percentage points) is strongly metric-specific. Q2 2024 is absent from the right-hand columns because its **decrease** cells are the seven missing values, the two data-quality issues touch exactly the same rows.

This is why nothing is rescaled: there is no single residual to redistribute, because its size depends on which metric is being looked at.

3.2 How are the percentages distributed?

Form: histogram.

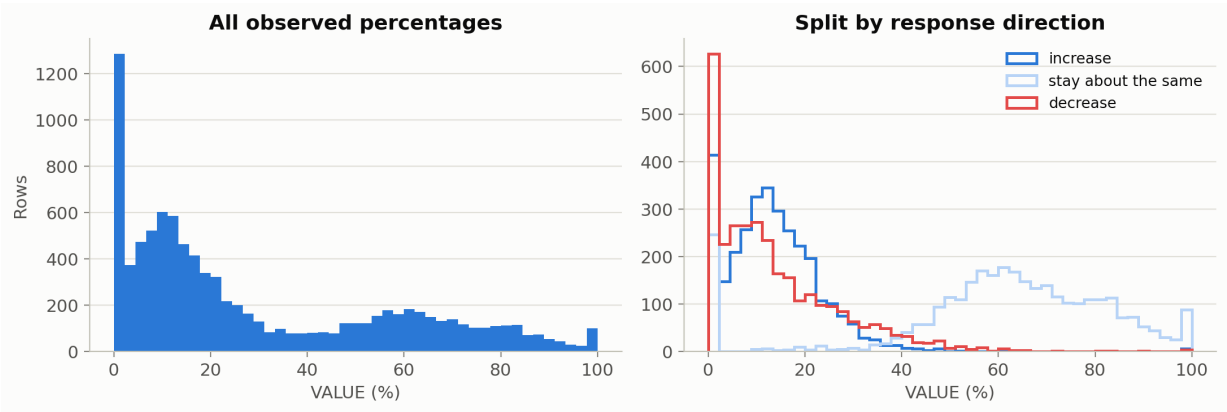


Figure 3: stay about the same sits far right of the other two: most businesses expect stability on most questions. Over a thousand rows read exactly 0%, mostly territories crossed with narrow industries, so group-level findings need a volatility caveat.

3.3 Do the four metrics move together?

The source data is *long*, so a correlation between raw columns would be meaningless (GEO against VALUE). The data is first reshaped to one row per survey cell and one column per metric; only then is the question computable. 645 cells have all four metrics observed and non-empty.

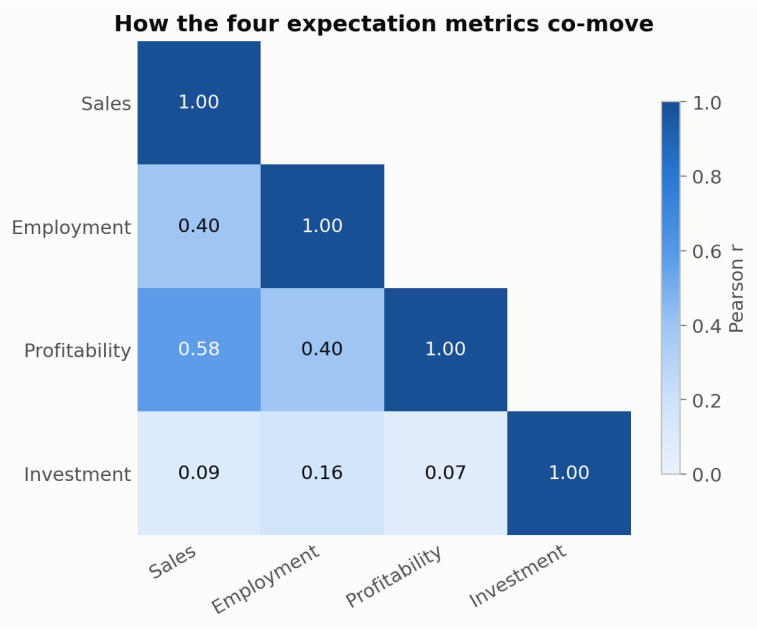


Figure 4: Pearson correlation between the four metrics' increase shares.

- Sales ↔ Profitability is the strongest pair ($r = 0.58$). Where businesses expect to

Metric	Mean r with the other three
Sales	0.36
Profitability	0.35
Employment	0.32
Investment	0.10

Table 5: Investment is close to independent of the rest.



Figure 5: The strongest pair, cell by cell. This is the relationship Section 5 finds inverted in Q2 2024.

sell more, they usually expect to earn more. This is the normal relationship, and it is what makes Section 5 interesting.

- **Employment sits in the middle** ($r \approx 0.40$ with both), consistent with hiring following expected activity.
- **Investment is nearly independent of all three** (mean $r = 0.10$). It is not a lagging version of the operating cycle.

What is done about it

Correlation changes the analysis rather than merely describing it:

1. **No composite index.** Averaging sales, employment and profitability into one “confidence” number would triple-count one shared underlying factor and present it as three independent confirmations. The metrics stay separate.
2. **Investment is analysed on its own terms.** It carries information the others do not, so folding it into an operating bundle would hide the only genuinely independent signal in the data.
3. **Sales and profitability are never treated as two independent pieces of evidence.** When they agree, that is close to one finding, not two.

3.4 Are the three response shares mechanically linked?

If the three directions were a closed split of every respondent, **increase** and **decrease** would have to trade off, strongly negatively correlated. They do not: r ranges from -0.10 to $+0.05$ across the four metrics.

The three shares are not a closed partition, so $100 - \text{increase} - \text{same}$ is not a valid reconstruction and the seven missing cells must stay missing.

This is the empirical basis for the preprocessing decision, rather than a stylistic preference.

4 Analysis 1: operating expectations strengthened, investment did not

Measure: the directly observed share expecting an *increase*, at the Canada / all-industries level. It is what the survey reports, it is comparable across quarters, and it needs no imputation.

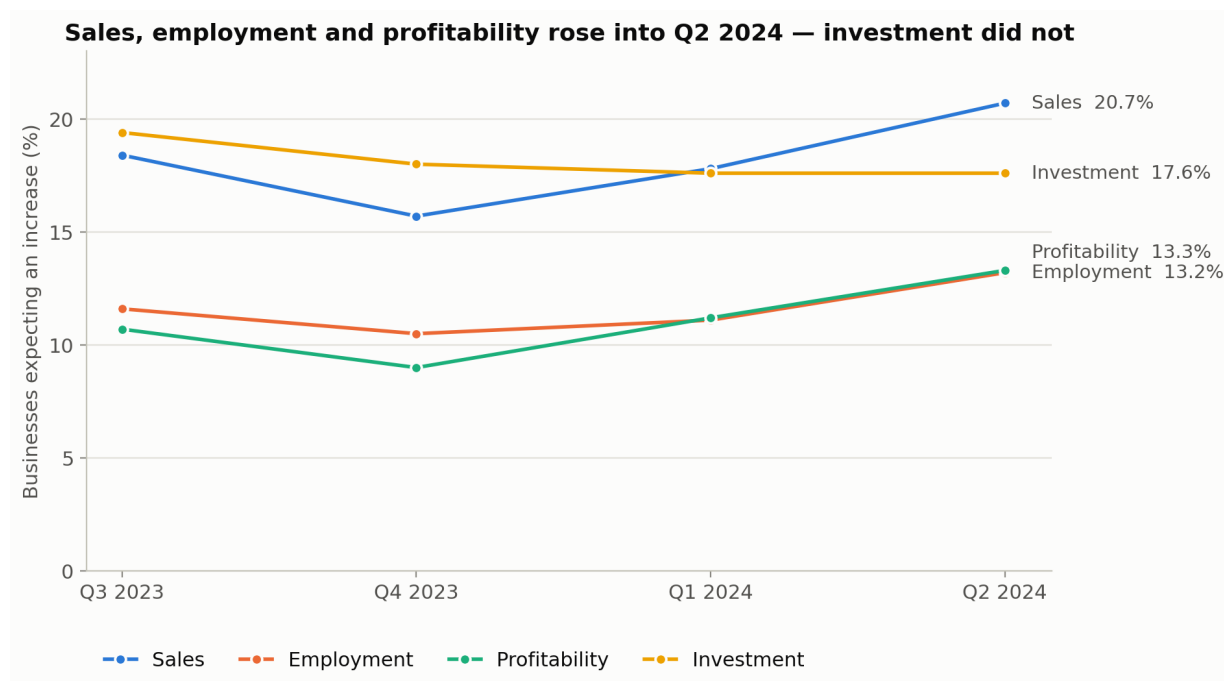


Figure 6: Share of businesses expecting an increase, by metric and quarter.

Metric	Q1 2024	Q2 2024	Change	4-quarter high?
Sales	17.8%	20.7%	+2.9 pp	yes
Employment	11.1%	13.2%	+2.1 pp	yes
Profitability	11.2%	13.3%	+2.1 pp	yes
Investment	17.6%	17.6%	0.0 pp	no

Table 6: Quarter-over-quarter change in the share expecting an increase.

Is comparing these four side by side fair?

Section 3.1 showed the metrics have different answer rates, which invites the objection that the comparison is not like-for-like. Two answers:

On the survey’s own definition it is valid. VALUE is the percentage of *all* business owners in the group, so every share already sits on the same denominator.

And the conclusion survives the alternative definition anyway. Rescaling each share to only those who gave a direction, investment’s increase share still *falls*: 22.9% (Q3 2023) → 21.2% → 21.0% (Q1 2024). Q2 2024 cannot be rescaled at all, because its **decrease** cells are the missing ones, which is itself a reason to prefer the raw observed share as the headline measure: it is the only one available for every quarter.

Interpretation

Sales, employment and profitability increase-expectations all reached their highest level in the four-quarter window in Q2 2024, on the face of it a broad improvement in near-term operating expectations. Investment is the exception: flat at 17.6% and still below its 19.4% Q3 2023 level.

Section 3.3 explains why that is not merely a lag. Investment expectations are statistically independent of the other three, and one business in six has no investment direction to report. Investment is a different decision, not a slower-moving version of the same one.

What I would tell a stakeholder. Q2 2024 is a genuine improvement in how businesses expect to trade over the next three months, but not yet a broad-based recovery in confidence, firms are more positive about activity than about committing capital. I would resist the natural narrative that investment will follow once sales improve: nothing here supports that link, and the correlation structure argues against it. Testing it properly needs financing conditions, capacity utilisation and credit availability, none of which this survey contains.

5 Analysis 2: profitability pressure is broad across industries

Measure:

Net profitability balance = % expecting an increase - % expecting a decrease

Stay about the same is retained as genuinely neutral, it is not split between the two sides. Critically, the balance requires **no assumption that the three shares total 100**, which Section 3.1 showed would be false.

Selected Q2 2024 industries	Increase	Decrease	Net balance
Agriculture, forestry, fishing and hunting	6.5%	47.4%	−40.9 pp
Accommodation and food services	14.2%	44.2%	−30.0 pp
Transportation and warehousing	13.1%	41.0%	−27.9 pp
Finance and insurance	18.7%	16.2%	+2.5 pp

Table 7: Finance and insurance is the only industry in positive territory.

Interpretation

In Q2 2024, **12 of 13 industries** carry a negative profitability balance. Expected margin pressure is broad, not sector-specific.

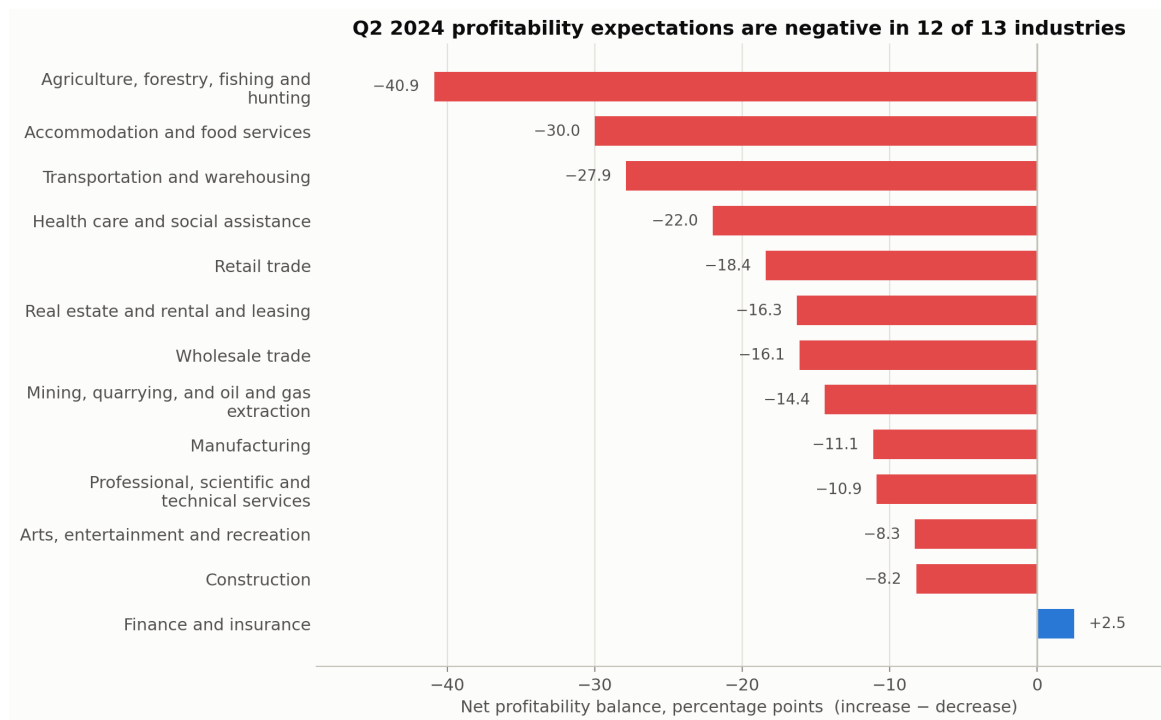


Figure 7: Q2 2024, Canada, by industry, sorted worst-first. A pie chart would be wrong here: 13 categories, many close together, and the values are not parts of a whole.

Is this new, or has it always been like this?

A single quarter cannot say, and it is the first question a stakeholder should ask, “margins are under pressure” means nothing without a baseline.

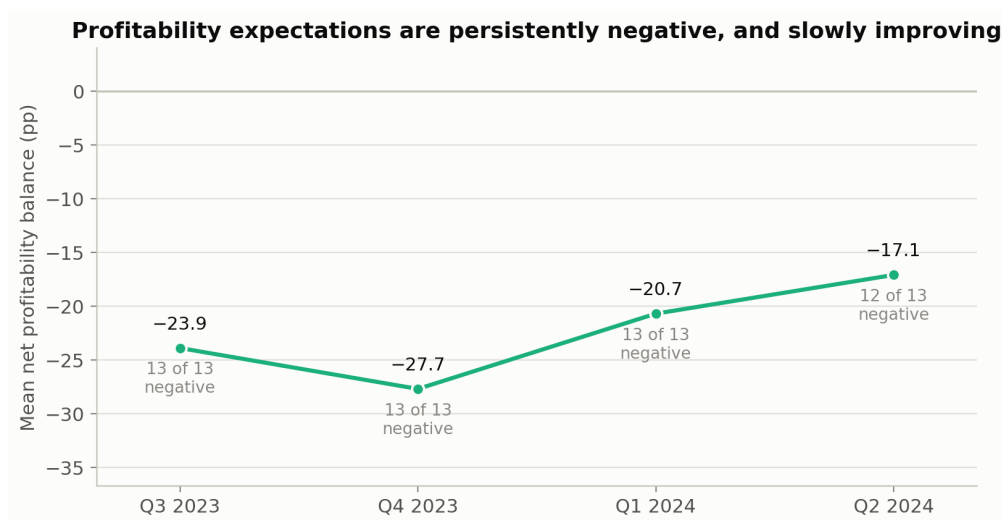


Figure 8: The same measure across all four quarters.

Margin pessimism is not new. Every one of the 13 industries carried a negative balance in each of the first three quarters. Q2 2024 is the least bad of the four: the mean improved from -23.9 pp to -17.1 pp and, for the first time, one industry turned positive.

So the honest framing is **entrenched, slowly easing pessimism**, not an emerging squeeze. Note too that profitability improved *alongside* sales rather than against it, which is what the $r = 0.58$

of Section 3.3 predicts. **The level is the finding, not a divergence in direction:** even at the best reading in the window, 12 of 13 industries still expect profitability to fall while sales expectations sit at a four-quarter high, and 8 of 13 expect sales growth of 15% or more *and* falling margins simultaneously.

That combination points away from a demand problem and toward a *cost* problem: firms anticipating they can sell more without keeping more of it. Input costs, wages, financing costs and limited pricing power are the obvious candidates, but this dataset contains none of them, so they remain hypotheses to test rather than conclusions to report.

What I would tell a stakeholder. Do not read the improving national sales figure as improving business health. The more decision-relevant signal is that margin expectations are negative almost everywhere. I would use this balance as a screening tool: agriculture, accommodation and food services, and transportation and warehousing are where deeper sector work would pay off first.

6 Analysis 3: the improvement was not shared evenly

Canada is only 1 of the 14 geographies in this extract. The other 13 account for roughly 93% of the rows and, unlike Q2 2024’s national cells, contain **no missing values at all**.

Measure: the change in the share expecting sales to increase, Q1 2024 → Q2 2024, per geography; reported alongside each jurisdiction’s four-quarter volatility, because the two must be read together.

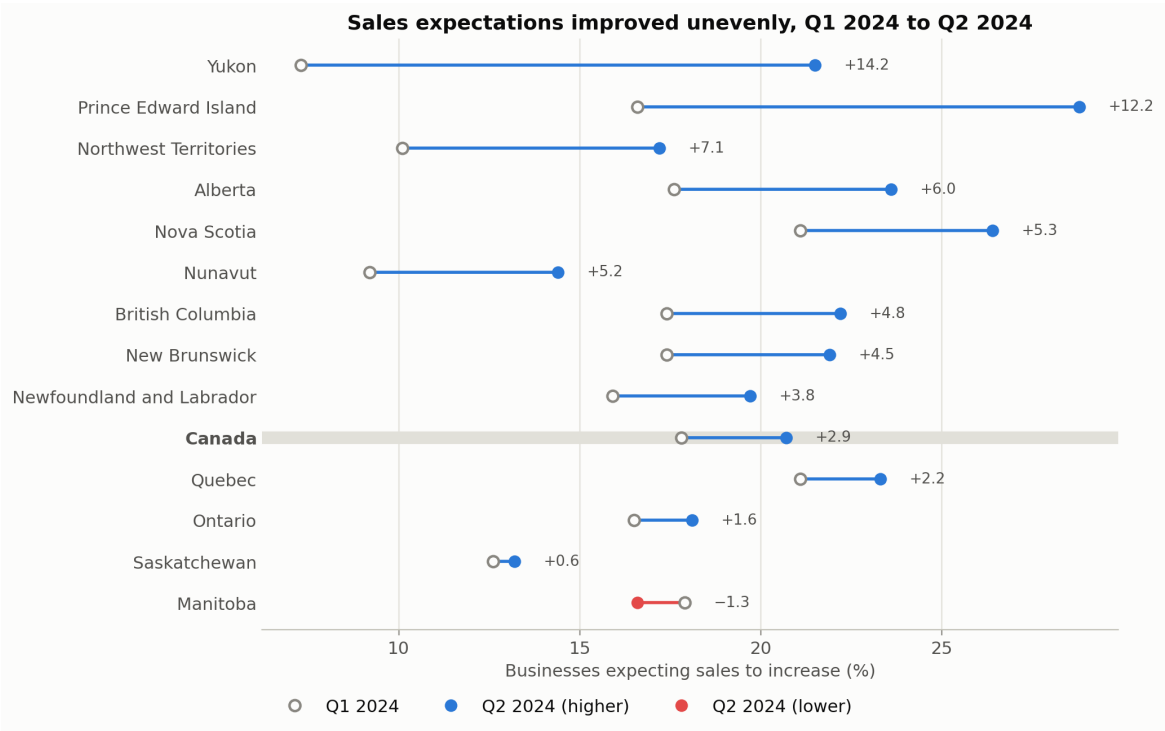


Figure 9: Dumbbell chart: the connector length *is* the change, and its colour the direction. A grouped bar chart would show the same numbers but force the reader to subtract. Hollow dot = Q1 2024, filled = Q2 2024, so time order survives without colour.

Jurisdiction	Q1 2024	Q2 2024	Change
Yukon	7.3%	21.5%	+14.2 pp
Prince Edward Island	16.6%	28.8%	+12.2 pp
Alberta	17.6%	23.6%	+6.0 pp
British Columbia	17.4%	22.2%	+4.8 pp
Canada	17.8%	20.7%	+2.9 pp
Quebec	21.1%	23.3%	+2.2 pp
Ontario	16.5%	18.1%	+1.6 pp
Saskatchewan	12.6%	13.2%	+0.6 pp
Manitoba	17.9%	16.6%	−1.3 pp

Table 8: Selected jurisdictions. The full table is in `outputs/analysis_3_metrics.csv`.

The volatility caveat

The eye is drawn to Yukon and Prince Edward Island. But the four smallest jurisdictions have a mean four-quarter volatility of **5.2 pp**, against **2.6 pp** for the four largest provinces.

A +14 pp move in Yukon is not evidence of a Yukon boom , it is close to ordinary noise for a jurisdiction that volatile. The large provinces’ smaller moves carry far more information per percentage point.

This is why the metric table ships a volatility column: the caveat has to travel with the number.

Interpretation

The national +2.9 pp improvement conceals a spread from −1.3 to +14.2 pp. Twelve of thirteen jurisdictions improved; **Manitoba was the only decline**, and Saskatchewan was effectively flat. Most importantly, **the two largest economies lagged the national figure**: Ontario +1.6 pp and Quebec +2.2 pp, both below the national +2.9, which is pulled up by Alberta, British Columbia and Atlantic Canada.

What I would tell a stakeholder. Be careful quoting the national figure as “how Canadian businesses feel”. A member in Ontario experienced roughly half the improvement the national number implies, and one in Manitoba experienced none. For regional programme decisions the provincial detail matters more than the national average, and any decision keyed to the territories should treat single-quarter movements there as provisional.

7 Observations and conclusions

The three analyses each answer one question. Read together they tell a single story, and it is not the one the national headline suggests.

Step 1: The top line improved

Three of four metrics reached four-quarter highs. Investment alone stayed flat. Read on its own, this looks like a recovery with one puzzling laggard.

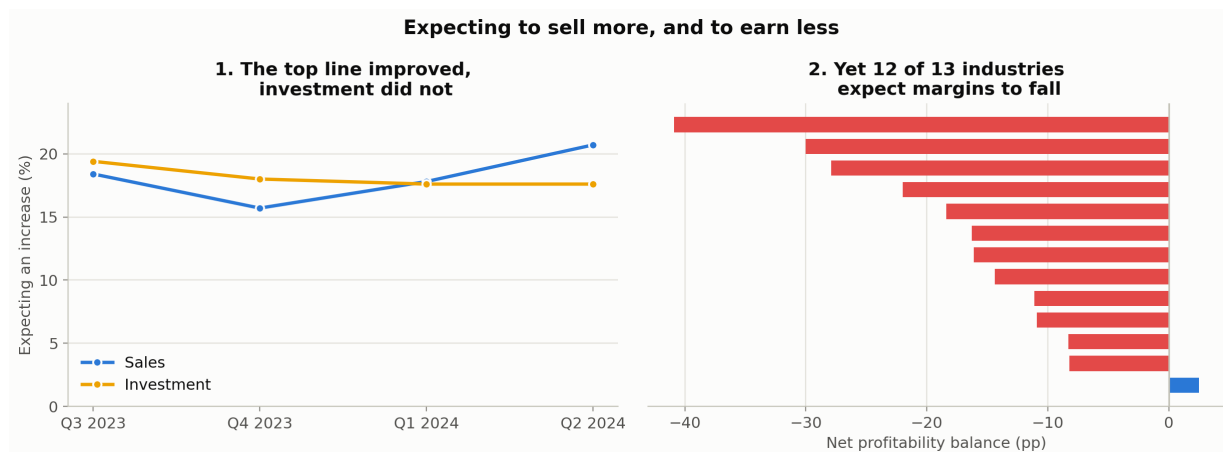


Figure 10: The whole argument in one figure.

Step 2: Investment is not simply a laggard

The intuitive story: capital spending follows sales with a delay, is contradicted on two independent measures: one business in six gives investment no directional answer, and investment is statistically independent of the other three metrics. If it were a lagged version of the operating cycle it would still be *correlated*, just offset. It is not.

Investment is a separate decision, made on different grounds, so its flatness is not a delay to wait out.

Step 3: Entrenched margin pessimism

Twelve of thirteen industries expect profitability to fall in Q2 2024, and every industry expected it in each of the three preceding quarters. The direction is encouraging; the mean balance improved from -23.9 pp to -17.1 pp, so profitability moved *with* sales, as the $r = 0.58$ would predict. The level is what matters: even at the best reading in the window, businesses expect to sell more and keep less, and have done so for a year.

That is a cost-side story rather than a demand-side one, and it explains Step 2 better than a sudden squeeze would: a firm that has expected thinner margins for four straight quarters has little reason to commit capital, whatever its sales outlook does.

Step 4: And it was not shared evenly

Ontario and Quebec both came in below the national change; Manitoba declined. Most Canadian businesses sit in the provinces that improved least.

What I would tell a stakeholder

In one line: business expectations improved in Q2 2024, but the improvement is narrower and shallower than the national figure suggests, firms still expect to sell more and earn less, and they are not planning to invest.

- 1. Read the level, not just the change.** Q2 2024 is genuinely the best of the four quarters on margin expectations, but 12 of 13 industries still expect profitability to fall, and every

industry expected it in every quarter of the window. A year of thin expected margins shapes behaviour more than one quarter of improvement does.

2. **Stop waiting for investment to follow.** The two are statistically unrelated here. If investment is a policy goal it needs addressing on its own terms.
3. **Use provincial detail, not the national average, for regional decisions.**

Given the cost-side reading, the next place to look is input costs, wage pressure, financing costs and pricing power, a hypothesis this survey generates but cannot test.

8 Part 2: the quarterly refresh pipeline

Implemented in `src/csbc.py`, demonstrated in `src/07_quarterly_refresh_pipeline.ipynb`, and runnable as `src/refresh_quarter.py`.

8.1 Two modes, because a new file answers two questions

Mode	Needs history?	Question it answers
standalone	no	What does this new quarter say on its own? Composition, data quality and sector balance for the quarter in isolation.
append	yes	How does the whole series look now? Appends the quarter, re-validates the union, refreshes the trend across all quarters.
both	yes	Both, in that order (default).

Table 9: The two analyses the pipeline provides.

The ordering matters: **standalone runs first**, so a new file can be inspected *before* any decision to admit it to the historical series. If it looks wrong, nothing has been written.

```
python src/refresh_quarter.py --new-file "data/raw/Data CSBC-Q3 2024.csv"
python src/refresh_quarter.py --new-file new.csv --mode standalone
python src/refresh_quarter.py --new-file new.csv --mode append
```

The script exits non-zero and writes nothing when a guard fires, so a scheduler can detect the failure.

8.2 Flow

```
new quarterly CSV
|
+--> schema / quarter / range / key / category validation
|
+--> known label normalisation
|
+--> MODE standalone: composition + sector balance for this quarter alone
|
+--> duplicate-quarter check
```

```

|
+--> append to combined history
|
+--> re-validate the union
|
+--> MODE append: refresh the trend chart across all quarters

```

8.3 What could break it, and how it is handled

The most likely future break is **schema or category drift**: a renamed column, or a new category value. The danger is not a crash: it is a *silent* split of a historical series, where the chart still renders and still looks plausible, so nobody reviews it.

The approach is therefore to **surface the change rather than guess**: stop the load, compare the new label against the source definition, add an explicit versioned mapping if it is legitimate (exactly as was done for Q2 2024’s two renames), then re-run. That trade-off favours data integrity over producing a chart from a structurally different dataset.

These guards are not merely described, the pipeline notebook triggers each one and shows it blocked:

Attempt	Result
Re-ingesting a quarter already in the history	blocked
An unrecognised Business_information label	blocked
A renamed VALUE column	blocked
A percentage outside 0–100	blocked
An incoming file containing two quarters	blocked

Table 10: Guards, exercised live rather than asserted.

The one thing it deliberately does *not* treat as fatal is a missing value, those are expected, and are reported and retained.

9 Limitations

- **No causality.** Every relationship here is an association across survey cells. The margin-squeeze reading is the most plausible explanation of the pattern, not a demonstrated mechanism.
- **No statistical significance.** The extract carries percentages but no sample sizes, weights, confidence intervals or margins of error, so nothing here is significance-tested. That is why the volatility column in Section 6 matters, it is the closest available proxy for how much to trust a movement.
- **Four quarters is not a trend.** It supports quarter-to-quarter comparison, not a cycle claim.
- **Expectations, not outcomes.** Whether these expectations were realised is outside this data.
- **Correlations are cross-sectional.** With four periods there are too few for a time-series correlation, so Section 3.3 describes how metrics vary together *across groups*, not how they lead or lag each other.

- **The seven missing values remain missing**, and every published figure avoids depending on them.

If this were operationalised

A formal versioned schema contract for incoming files; structured logging and alerting on validation failures rather than a message on stderr; regression tests on the survey taxonomy so a renamed industry is caught before it reaches a chart; and above all obtaining the survey weighting and suppression metadata, which is the main ceiling on what this analysis can claim.

A Reproducibility

A.1 Layout

```
business_data_lab_assignment/
  README.md
  requirements.txt
  data/raw/                4 supplied quarterly CSVs
  data/processed/          combined_csbc.csv (written by notebook 01)
  outputs/                 all figures, metric tables, quality report
  report/                  this document
  src/csbc.py              shared: schema, validation, charts, pipeline
  src/refresh_quarter.py   pipeline CLI
  src/01..07 *.ipynb       the analysis, in run order
  tests/test_csbc.py       15 tests
```

A.2 Running it

Nothing needs to be run to review the submission, every notebook ships with its outputs saved. To re-run:

```
python -m venv .venv
.venv/bin/python -m pip install -r requirements.txt # Windows: .venv\Scripts\python.exe
.venv/bin/python -m pytest -q                     # expect 15 passed
.venv/bin/python -m jupyter lab
```

Then run notebooks 01–07 in order with *Restart Kernel and Run All Cells*.

All paths are derived from the location of `src/csbc.py`, never from the working directory, so the project runs from wherever it is unzipped and the notebooks work whether Jupyter is started in the project root or in `src/`.

A.3 Figure provenance

Every figure in this report is generated by the notebooks into `outputs/` and included from there, so the report cannot drift out of step with the analysis.